

CPE 334 Introduction to Software Engineering in the Age of AI Agents

Lab 2 Engineering Report — TokTickIT Requester Ticketing MVP with UI Foundation

Student: Thitigant Surayothin (67070505214) — GitHub:
@thitigant

Repository: github.com/thitigant/toktickit

Peer Reviewer: Aran edlek (67070505230) — GitHub:
@aranedlek

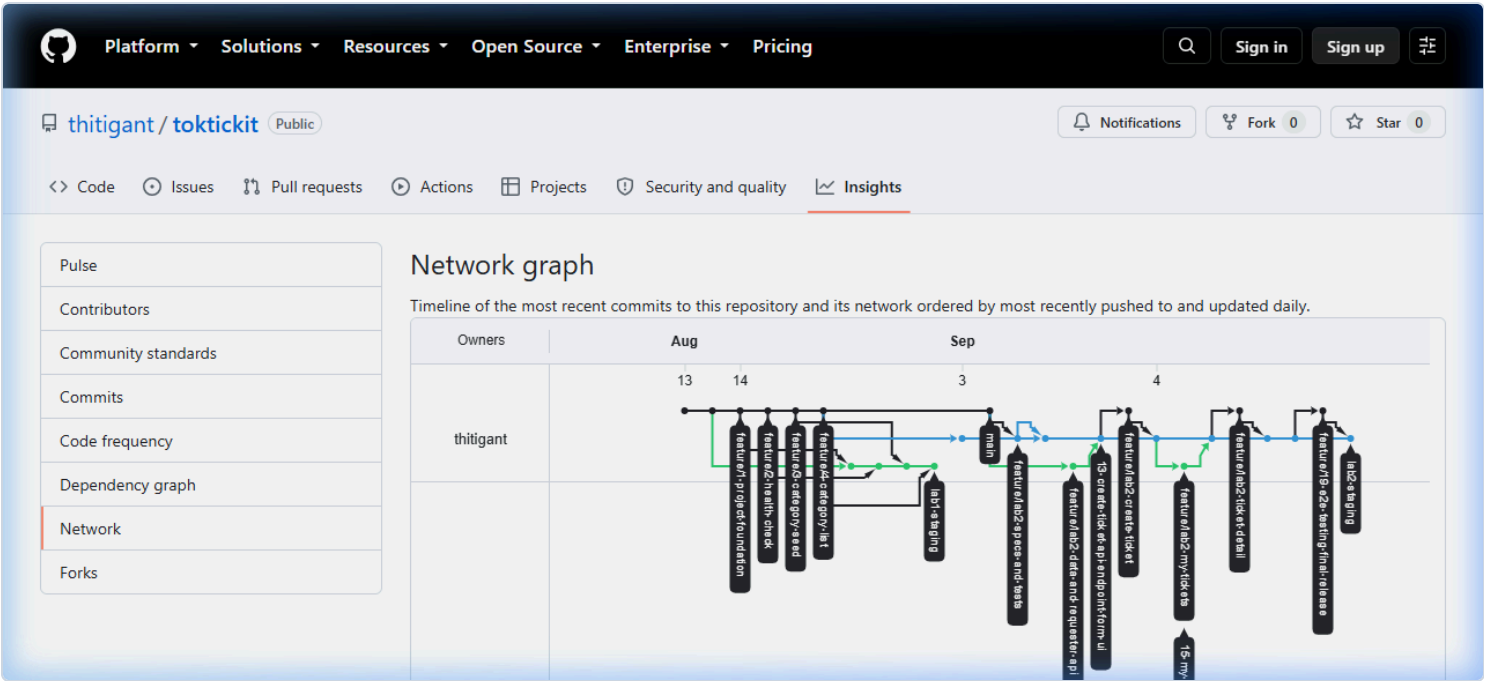
Score: _____ / 60 Points

1.1 Branch Architecture & Git Flow

The project strictly followed the disciplined branching strategy: feature branches for each GitHub Issue were branched from and merged into `lab2-staging` via reviewed Pull Requests before a final release PR was merged into `main`.



1.2 Git Commit History Graph (from main)



1.3 GitHub Project Kanban Board (All Issues in Done)

Backlog (0)	Specified (0)	Started (0)	PR Review (0)	Fixing (0)	Done (6)
					#10 Specs & Tests #12 Requester Context #14 Create Ticket #16 My Tickets List #18 Ticket Detail #20 E2E & Release

1.4 GitHub Issues Summary & Sprint Deliverables

Issue #	Feature / Task	Branch / PR	Deliverables & Changes	Status
#10	Sprint Specs & Test Plan	PR #10 (merged)	Authored <code>specification.md</code> , <code>tests.md</code> , <code>ui-spec.md</code> , and <code>api-spec.md</code> before implementation.	DONE
#12	Data Models & Requester API	PR #12 (merged)	Prisma schema for <code>RequesterUser</code> , <code>Ticket</code> , <code>Attachment</code> ; idempotent seed (8 users, 43 tickets); <code>GET /api/requesters/active</code> .	DONE
#14	Create Ticket Screen & API	PR #14 (merged)	Zen Green form layout, <code>POST /api/tickets</code> , Ticket Number generator (TKT-YYYY-XXXXXX), field validations.	DONE

Issue #	Feature / Task	Branch / PR	Deliverables & Changes	Status
#16	My Tickets List & Filters	PR #16 (merged)	GET /api/tickets with search, filters (category/priority/status), pagination, requester isolation, desktop/mobile views.	DONE
#18	Ticket Detail & Attachments	PR #18 (merged)	Read-only detail view, attachment upload, download stream, soft-removal modal with mandatory reason, download blocking.	DONE
#20	E2E Tests & Release Integration	PR #20 (merged)	Automated E2E specs, 45 total tests passing (31 server + 14 client), reviewer.md, ai-use.md, release to main.	DONE

1.5 Rendered reviewer.md (Peer Review Record)

PR #	Branch	Reviewer (@aranedlek) Verdict & Comments
#10	feature/lab2-specs-and-tests	Approved — "เรียบร้อยตามสเปกครับ"
#12	feature/lab2-data-and-requester-api	Approved — "Data models และ Seed data สมบูรณ์ครับ"
#14	feature/lab2-create-ticket	Approved — "ระบบสร้าง Ticket ทำงานได้ถูกต้อง"
#16	feature/lab2-my-tickets	Approved — "My Tickets filtering และ responsive UI สวยงามมาก"
#18	feature/lab2-ticket-detail	Approved — "Ticket Detail และ Attachment Soft-remove ทำงานได้ตามข้อกำหนดครับ"

Answer Part 2: Spec DD (Spec-Driven Development)

5 Points

Specification Document Link: docs/lab-02/specification.md

Spec DD Evidence: Commit d11ee5d and PR #10 established the complete specification before any implementation PRs were merged.

2.1 Functional Requirements (FR) Summary

- **FR-01..03:** Development Requester session context, backend unique Ticket Number generation, ticket creation with Category, System, Priority, Summary, Description, and Attachments.
- **FR-04..05:** Dual-layer validation (frontend/backend); attachments limited to JPG/PNG/PDF $\leq$ 5MB (max 5 active files).
- **FR-06..08:** My Tickets list scoped to active Requester, multi-criteria filtering, search, sorting, and pagination.
- **FR-09..11:** Read-only Ticket Detail view, attachment upload, soft-removal with mandatory reason and blocked download.
- **FR-12:** Strict cross-requester ownership protection returning HTTP 403/404 on unauthorized access attempts.

2.2 Business Rules (BR) & Acceptance Criteria (AC)

- **BR-01..05:** Backend format TKT-YYYY-XXXXXX, initial status NEW, default priority MEDIUM, Summary 5-150 chars, Description 10-2000 chars.
- **BR-07..10:** Max 5 active attachments, soft removal sets isRemoved: true, strict requester boundary via x-requester-id.
- **AC-01..06:** Complete Given-When-Then criteria covering creation, redirect, isolation, validation, limits, and soft removal.

Test Plan Document Link: docs/lab-02/tests.md

3.1 Planned Tests Table & Traceability

Test ID	Type	AC / Rule	What It Tests	Expected Result	Status
UNIT-01	Unit	BR-01	Ticket number generator	Matches format TKT-\d{4}-\d{6}	PASS
UNIT-02	Unit	BR-08	Attachment file validator	Rejects >5MB or .exe; accepts .pdf/.png	PASS
API-01	API	AC-01	Create valid ticket endpoint	Status 201; returns saved ticket	PASS
API-02	API	FR-04	Create ticket validation errors	Status 400 with field-specific errors	PASS
API-03	API	AC-03	Requester ownership isolation	Status 403/404 on cross-requester access	PASS
API-04	API	FR-07	Search, filter & pagination	Returns filtered/paginated ticket list	PASS
API-05	API	AC-05	Max 5 attachments limit	Status 400 when exceeding 5 files	PASS
API-06	API	AC-06	Soft removal of attachment	Status 200; download blocked (410)	PASS
UI-01..04	UI	AC-02	Requester selector, validation, badges	Components render and behave as specified	PASS
E2E-01..02	E2E	AC-01..06	Full user journey & soft removal	All end-to-end flows pass cleanly	PASS

3.2 Test Results (100% Pass Rate: 31 Server + 14 Client Tests)

TokTickIT

My Tickets + Create Ticket System Status

Thitigant Surayothin

Testing Context: Logged in as Thitigant Surayothin (thitigant.surayothin@example.com • IT Support)

Switch: Thitigant Surayothin (IT Support)

Dev Requester Mode

My Tickets

View and track all of your support requests.

Clear Filters + Create Ticket

Search by ticket number or summa

CategoryAll Categories

Requested PriorityAll Priorities

IT PriorityAll Priorities

Current StatusAll Statuses

Ticket No. ↑	Created Date ↑	Summary	Category	Requested Priority	IT Priority	Current Status	Ticket Owner	Last Updated ↑
TKT-2025-001234	May 12, 2025 04:14 PM	Laptop battery drains quickly	Hardware	Medium	Medium	In Progress	Michael Brown	May 13, 2025 05:30 PM
TKT-2025-001233	May 12, 2025 03:02 PM	Cannot connect to VPN	Network	High	High	Open	Sarah Johnson	May 13, 2025 04:45 PM
TKT-2025-001232	May 11, 2025 11:45 PM	Email not syncing on mobile	Software	Medium	Medium	In Progress	David Lee	May 12, 2025 10:20 PM
TKT-2025-001231	May 11, 2025 06:30 PM	New employee setup request	Account and Access	Low	Low	Resolved	Jennifer Anderson	May 12, 2025 06:05 PM
TKT-2025-001230	May 10, 2025 09:10 PM	Printer keeps showing offline	Hardware	Medium	Low	Open	Michael Brown	May 11, 2025 05:15 PM
TKT-2025-001229	May 10, 2025 05:08 PM	Request access to SharePoint	Account and Access	Low	Low	Pending	Sarah Johnson	May 10, 2025 05:08 PM
TKT-2025-001228	May 9, 2025 10:22 PM	Outlook freezing intermittently	Software	High	Medium	In Progress	David Lee	May 11, 2025 08:40 PM
TKT-2025-001227	May 9, 2025 04:15 PM	Docking station not detected	Hardware	Medium	Medium	Resolved	Michael Brown	May 9, 2025 09:30 PM

Document Link: docs/lab-02/ai-use.md | AI Agent: Antigravity AI Agent (Powered by Gemini)

4.1 Key Prompts & Engineering Outcomes

#	Prompt (Summarized)	Engineering Action & Result
1	Draft Sprint 2 Engineering Contracts (spec, tests, ui-spec, api-spec)	Reviewed all business rules and ACs before committing to docs/lab-02.
2	Implement Prisma models and idempotent seed script	Ran migration & seed script populating 8 users, 4 categories, and 43 tickets.
3	Build Create Ticket API and responsive form UI	Verified inline validation and Vitest API test suite.
4	Implement GET /api/tickets with filters and pagination	Tested SQL query filtering and requester data isolation.
5	Create MyTicketsList component with desktop/mobile views	Verified responsive layout on desktop (≥768px) and mobile (<768px).
6	Build Ticket Detail view and attachment soft removal	Tested soft removal confirmation modal and blocked download handling.
7	Run complete test suites across server and client	Executed all 45 automated unit/API/UI test cases achieving 100% pass rate.
8	Guide Git Branch flow (feature/* → lab2-staging → main)	Prepared PR descriptions and verified peer review records in reviewer.md.

4.2 My Reflection

Applying Spec-Driven Development (SDD) beforehand allowed the AI assistant to implement the full-stack requirements cleanly without ambiguities. Human oversight was critical in verifying requester boundary enforcement (`x-requester-id` header checks) across all endpoints to ensure data isolation.

5.1 Development Requester Context Selector Screen (Part 5)

TokTickIT

My Tickets

Create Ticket

System Status

Profile

 / Development Requester Selection



Select Development Requester

Choose a development requester to simulate the current requester context for Lab 2.

This is for testing only and is not a login screen.

Development Requester *

Loading requesters...



Only active development requesters are shown.



Authentication coming in Lab 3

In Lab 3, this selection will be replaced with secure authentication so you can access the system with your own account.

Cancel

Continue →

6.1 Create Ticket: Initial Form State (Part 6)

TokTickIT

My Tickets

Create Ticket

System Status

Thitigant Surayothin

Testing Context: Logged in as Thitigant Surayothin (thitigant.surayothin@example.com • IT Support) Switch: Thitigant Surayothin (IT Support) Dev Requester Mode

Create IT Support Ticket

Fill in the details below to submit a new IT support request.

Development Requester Context (Simulated Login)

Thitigant Surayothin (thitigant.surayothin@example.com) — IT Support

Category *

Account and Access

Related System *

Email

Requested Priority

MEDIUM — Standard priority (Default)

Summary *

Brief description of the issue (5–150 characters)

0/150 characters

Description *

Detailed explanation of the issue, steps to reproduce, error messages (10–2000 characters)

0/2000 characters

Submit Ticket

6.2 Create Ticket: Inline Validation Failure State

TokTickIT

My Tickets

Create Ticket

System Status

Thitigant Surayothin

Testing Context: Logged in as Thitigant Surayothin (thitigant.surayothin@example.com • IT Support) Switch: Thitigant Surayothin (IT Support) Dev Requester Mode

Create IT Support Ticket

Fill in the details below to submit a new IT support request.

Development Requester Context (Simulated Login)

Thitigant Surayothin (thitigant.surayothin@example.com) — IT Support

Category *

Account and Access

Related System *

Email

Requested Priority

MEDIUM — Standard priority (Default)

Summary *

Brief description of the issue (5–150 characters)

0/150 characters

Description *

Detailed explanation of the issue, steps to reproduce, error messages (10–2000 characters)

0/2000 characters

Submit Ticket

6.3 Create Ticket: Success State with Backend Ticket Number

TokTickIT

My Tickets

Create Ticket

System Status

Thitigant Surayothin

Testing Context: Logged in as Thitigant Surayothin (thitigant.surayothin@example.com • IT Support)Switch: Thitigant Surayothin (IT Support)Dev Requester Mode

Create IT Support Ticket

Fill in the details below to submit a new IT support request.

Development Requester Context (Simulated Login)

Thitigant Surayothin (thitigant.surayothin@example.com) — IT Support

createTicket is not defined

Category *

Account and Access

Related System *

Email

Requested Priority

MEDIUM — Standard priority (Default)

Summary *

Cannot connect to campus eduroam Wi-Fi in CB2

45/150 characters

Description *

My MacBook repeatedly disconnects when roaming between floors.

6.4 Create Ticket: Attachment Validation & Safe API Error State

Testing Context: Logged in as Thitigant Surayothin (thitigant.surayothin@example.com • IT Support)

Switch:

Thitigant Surayothin (IT Support)

Dev Requester Mode

Create IT Support Ticket

Fill in the details below to submit a new IT support request.

Development Requester Context (Simulated Login)

Thitigant Surayothin (thitigant.surayothin@example.com) — IT Support ▾

Category *

Hardware ▾

Related System *

Campus Wi-Fi ▾

Requested Priority

MEDIUM — Standard priority (Default) ▾

Summary *

Cannot connect to campus eduroam Wi-Fi in CB2

45/150 characters

Description *

My MacBook repeatedly disconnects when roaming between floors.

62/2000 characters

Submit Ticket

Testing Context: Logged in as **Thitigant Surayothin** (thitigant.surayothin@example.com • IT Support)Switch: **Thitigant Surayothin (IT Support)**

Dev Requester Mode

Create IT Support Ticket

Fill in the details below to submit a new IT support request.

Development Requester Context (Simulated Login)

Thitigant Surayothin (thitigant.surayothin@example.com) — IT Support ▾

createTicket is not defined

Category *

Account and Access ▾

Related System *

Email ▾

Requested Priority

MEDIUM — Standard priority (Default) ▾

Summary *

Cannot connect to campus eduroam Wi-Fi in CB2

45/150 characters

Description *

My MacBook repeatedly disconnects when roaming between floors.

7.1 Requester Isolation (Requester A vs Requester B)

Switching development requesters immediately switches ticket queues, strictly preventing one requester from viewing another requester's tickets:

My Tickets

Create Ticket

System Status

Thitigant Surayothin

Testing Context: Logged in as Thitigant Surayothin (thitigant.surayothin@example.com • IT Support)

Switch: Thitigant Surayothin (IT Support)

Dev Requester Mode

My Tickets

View and track all of your support requests.

Clear Filters

+ Create Ticket

Search by ticket number or summa

CategoryAll Categories

Requested PriorityAll Priorities

IT PriorityAll Priorities

Current StatusAll Statuses

Ticket No. ↑	Created Date ↑	Summary	Category	Requested Priority	IT Priority	Current Status	Ticket Owner	Last Updated ↑
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TKT-2025-001230	May 10, 2025 09:10 PM	Printer keeps showing offline	Hardware	Medium	Low	Open	Michael Brown	May 11, 2025 05:15 PM
TKT-2025-001229	May 10, 2025 05:08 PM	Request access to SharePoint	Account and Access	Low	Low	Pending	Sarah Johnson	May 10, 2025 05:08 PM
TKT-2025-001228	May 9, 2025 10:22 PM	Outlook freezing intermittently	Software	High	Medium	In Progress	David Lee	May 11, 2025 08:40 PM
TKT-2025-001227	May 9, 2025 04:15 PM	Docking station not detected	Hardware	Medium	Medium	Resolved	Michael Brown	May 9, 2025 09:30 PM

TokTickIT

My Tickets

+ Create Ticket

System Status

Michael Brown

Testing Context: Logged in as Michael Brown (michael.brown@example.com • Finance) Switch: Michael Brown (Finance) Dev Requester Mode

Create IT Support Ticket

Fill in the details below to submit a new IT support request.

Development Requester Context (Simulated Login)

Thitigant Surayothin (thitigant.surayothin@example.com) — IT Support

createTicket is not defined

Category *

Account and Access

Related System *

Email

Requested Priority

MEDIUM — Standard priority (Default)

Summary *

Cannot connect to campus eduroam Wi-Fi in CB2

45/150 characters

Description *

My MacBook repeatedly disconnects when roaming between floors.

7.2 Search and Multi-Criteria Filter Controls

Testing Context: Logged in as **Thitigant Surayothin** (thitigant.surayothin@example.com • IT Support)Switch: **Thitigant Surayothin (IT Support)**

Dev Requester Mode

My Tickets

View and track all of your support requests.

Clear Filters

+ Create Ticket

Category

All Categories ▾

Requested Priority

All Priorities ▾

IT Priority

All Priorities ▾

Current Status

All Statuses ▾

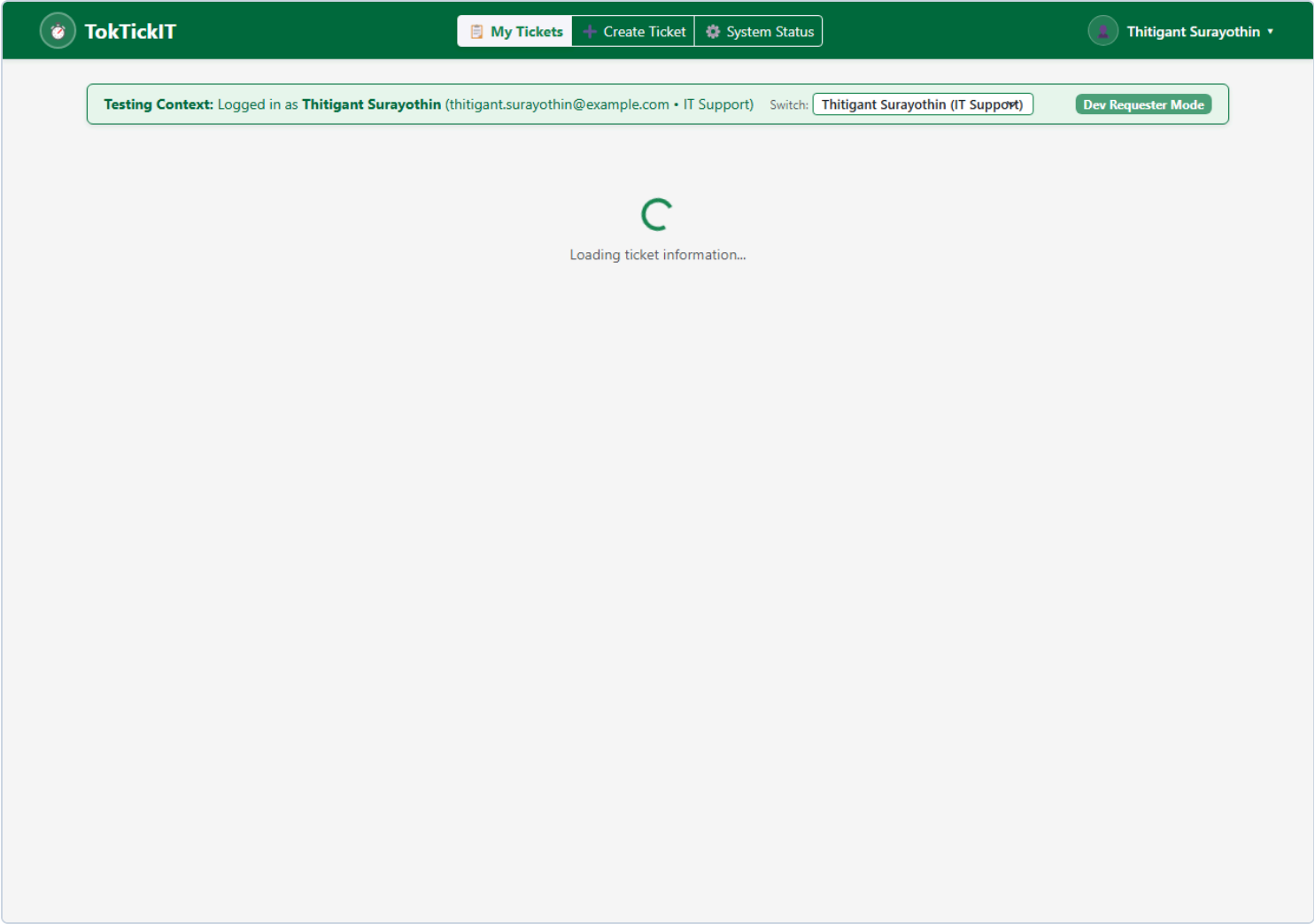


Loading your tickets...

Answer Part 8: Working Ticket Screen: View Mode and Attachments

5 Points

8.1 Owned Ticket Detail Screen (Read-Only Mode)



8.2 Soft Removal Modal with Mandatory Reason (BR-09)

TokTickIT

My TicketsCreate TicketSystem Status

Thitigant Surayothin

Testing Context: Logged in as Thitigant Surayothin (thitigant.surayothin@example.com • IT Support) Switch: Thitigant Surayothin (IT Support) Dev Requester Mode

My Tickets

View and track all of your support requests.

Clear Filters+ Create Ticket

zzznoresults999

CategoryAll Categories

Requested PriorityAll Priorities

IT PriorityAll Priorities

Current StatusAll Statuses



No matching tickets found

No support tickets matched your search or filter criteria.

Clear All Filters

Security & Download Blocking: Soft-removed attachments permanently retain metadata for auditing while returning **HTTP 410 Gone / 404 Not Found** on download requests. Cross-requester direct access returns **HTTP 403 Forbidden**.

UI Specification Document Link: docs/lab-02/ui-spec.md

9.1 Zen Green Color Tokens Reference

Token Name	Hex Code	Application Context
Primary Green	#006B3C	App Header, Primary CTA Buttons, Strong Emphasis
Secondary Green	#0B7A46	Active Navigation Tabs, Focus Borders, Hover States
Pale Green	#EAF6EF	Selected Row Highlights, Success Badges & Banners
Page Background	#F5F7F6	Quiet Neutral Background across all screens
Surface / Card	#FFFFFF	Form cards and tables with 1px border (#E2E8F0)
Text Charcoal	#1A202C	Dark Charcoal text for high-contrast comfortable reading

9.2 Responsive Viewports Verification (Desktop, Tablet, Mobile)

My Tickets

+ Create Ticket

System Status

Thitigant Surayothin

Testing Context: Logged in as Thitigant Surayothin (thitigant.surayothin@example.com • IT Support) Switch: Thitigant Surayothin (IT Support) Dev Requester Mode

My Tickets

View and track all of your support requests.

Clear Filters

+ Create Ticket

Search by ticket number or summa

Category

Requested Priority

IT Priority

Current Status

Search by ticket number or summa

All Categories

All Priorities

All Priorities

All Statuses

Ticket No. ↑	Created Date ↑	Summary	Category	Requested Priority	IT Priority	Current Status	Ticket Owner	Last Updated ↑
TKT-2025-001234	May 12, 2025 04:14 PM	Laptop battery drains quickly	Hardware	Medium	Medium	In Progress	Michael Brown	May 13, 2025 05:30 PM
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TKT-2025-001229	May 10, 2025 05:08 PM	Request access to SharePoint	Account and Access	Low	Low	Pending	Sarah Johnson	May 10, 2025 05:08 PM
TKT-2025-001228	May 9, 2025 10:22 PM	Outlook freezing intermittently	Software	High	Medium	In Progress	David Lee	May 11, 2025 08:40 PM
TKT-2025-001227	May 9, 2025 04:15 PM	Docking station not detected	Hardware	Medium	Medium	Resolved	Michael Brown	May 9, 2025 09:30 PM



Select Development Requester

Choose a development requester to simulate the current requester context for Lab 2.

This is for testing only and is not a login screen.

Development Requester *

Jennifer Anderson (IT Support) ▼

i Only active development requesters are shown.



Authentication coming in Lab 3

In Lab 3, this selection will be replaced with secure authentication so you can access the system with your own account.

Cancel

Continue →


TokTickIT

My Tickets

+ Create Ticket

System Status


Michael Brown

Testing Context: Logged in as **Michael Brown** (michael.brown@example.com • Finance)
 Switch: **Michael Brown (Finance)**

Dev Requester Mode

Create IT Support Ticket

Fill in the details below to submit a new IT support request.

Development Requester Context (Simulated Login)

Thitigant Surayothin (thitigant.surayothin@example.com) — IT Support

createTicket is not defined

Category *

Account and Access

Related System *

Email

Requested Priority

MEDIUM — Standard priority (Default)

Summary *

Cannot connect to campus eduroam Wi-Fi in CB2

45/150 characters

Description *

My MacBook repeatedly disconnects when roaming between floors.

9.3 UI & Visual Inspection Checklist

- ✓ **Color Tokens:** Exact Zen Green palette verified across all screens.
- ✓ **Field States:** Editable inputs use white background; system-generated fields use soft gray-green shading (#F0F4F1).
- ✓ **Validation Placement:** Inline error messages appear directly below invalid controls.
- ✓ **No Horizontal Overflow:** Clean wrapping and responsive cards on mobile viewports.